

2η Σειρά Ασκήσεων
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Άσκηση 1

```
IN 10H
LXI H,0900H
MVI A,FFH           ;A = 255
LXI D,0000H         ;WE COUNT THE ZEROS WITH D
MVI C,00H

FOR:
    CPI 00H
    JZ PUT_ZERO
    MOV M,A
    DCR A
    INX H
    JMP FOR

PUT_ZERO:           ;TO PUT ZERO IN THE LAST POSITION
    MOV M,A

MVI A,00H           ;A = 0 ,IN ORDER TO CHECK
CHECK:
    CPI FFH
    JZ EXIT
    CMP M
    JNZ ERROR
    CPI 20H
    JM A1
    CPI 71H
    JP A1
    INR C

A1:    DCX H
    INR A
    JMP CHECK
```

EXIT:

```
CMP M
JNZ ERROR
CMA
;STA 3000H      ;IF ALL OK -> 8 LEDS WILL OPEN
CMA
PUSH B          ;WE SAVE C FOR THE COUNTER
MVI C,00H
JMP COUNTER
```

ERROR:

```
MVI A,FEH
;STA 3000H      ;IF THERE IS AN ERROR THE 1ST LED WILL OPEN

JMP END
```

COUNTER:

```
                                ;A = 0 AND HL = 0900
MOV A,C                      ;C -> COUNTER
CALL COUNT_ZEROS
INX H
INR C
MOV A,C
CPI 00H
JZ DISPLAY
JMP COUNTER
```

DISPLAY:

```
POP B                        ;FROM COUNTER
MOV A,D                      ;HERE YOU CAN PUT D,E OR C , IN ORDER TO
                                ;SEE THE RESULTS
CMA                          ;WE EXPECT 1024D = 000001000000000000 B
STA 3000H                    ;D = 00000100 , E = 00000000 , C = 0101000
JMP END
```

```

COUNT_ZEROS:  PUSH PSW          ;WE SAVE A AND FLAGS
                PUSH B           ;WE SAVE B,C
                MVI B,00H        ;WE CAN CHANGE A,B AND C IN THE ROUTINE
                                ;B IS THE COUNTER FOR THE 8 BITS

                MOV C,M          ;C = M = ((H) (L)) FROM COUNTER
START:          MOV A,B
                CPI 08H
                JZ RETURN

                MOV A,C
                RAR
                MOV C,A          ;WE UPDATE C ,BECAUSE WE RIGTH SHIFTED A
                JC CONT
                INX D
CONT:           INR B
                JMP START
RETURN:         POP B
                POP PSW
                RET

BETWEEN:       PUSH PSW

                MOV A,M
                CPI 20H
                |

END:           END

```

Για το ερώτημα (α) παρακάτω φαίνονται τα περιεχόμενα των θέσεων μνήμης
0900 H - 0900 H

0900	FF	0901	FE	0902	FD	0903	FC	0904	FB	0905	FA	0906	F9	0907	F8
0908	F7	0909	F6	090A	F5	090B	F4	090C	F3	090D	F2	090E	F1	090F	F0
0910	EF	0911	EE	0912	ED	0913	EC	0914	EB	0915	EA	0916	E9	0917	E8
0918	E7	0919	E6	091A	E5	091B	E4	091C	E3	091D	E2	091E	E1	091F	E0
0920	DF	0921	DE	0922	DD	0923	DC	0924	DB	0925	DA	0926	D9	0927	D8
0928	D7	0929	D6	092A	D5	092B	D4	092C	D3	092D	D2	092E	D1	092F	D0
0930	CF	0931	CE	0932	CD	0933	CC	0934	CB	0935	CA	0936	C9	0937	C8
0938	C7	0939	C6	093A	C5	093B	C4	093C	C3	093D	C2	093E	C1	093F	C0
0940	BF	0941	BE	0942	BD	0943	BC	0944	BB	0945	BA	0946	B9	0947	B8
0948	B7	0949	B6	094A	B5	094B	B4	094C	B3	094D	B2	094E	B1	094F	B0
0950	AF	0951	AE	0952	AD	0953	AC	0954	AB	0955	AA	0956	A9	0957	A8
0958	A7	0959	A6	095A	A5	095B	A4	095C	A3	095D	A2	095E	A1	095F	A0
0960	9F	0961	9E	0962	9D	0963	9C	0964	9B	0965	9A	0966	99	0967	98
0968	97	0969	96	096A	95	096B	94	096C	93	096D	92	096E	91	096F	90
0970	8F	0971	8E	0972	8D	0973	8C	0974	8B	0975	8A	0976	89	0977	88
0978	87	0979	86	097A	85	097B	84	097C	83	097D	82	097E	81	097F	80
0980	7F	0981	7E	0982	7D	0983	7C	0984	7B	0985	7A	0986	79	0987	78
0988	77	0989	76	098A	75	098B	74	098C	73	098D	72	098E	71	098F	70
0990	6F	0991	6E	0992	6D	0993	6C	0994	6B	0995	6A	0996	69	0997	68
0998	67	0999	66	099A	65	099B	64	099C	63	099D	62	099E	61	099F	60
09A0	5F	09A1	5E	09A2	5D	09A3	5C	09A4	5B	09A5	5A	09A6	59	09A7	58
09A8	57	09A9	56	09AA	55	09AB	54	09AC	53	09AD	52	09AE	51	09AF	50
09B0	4F	09B1	4E	09B2	4D	09B3	4C	09B4	4B	09B5	4A	09B6	49	09B7	48
09B8	47	09B9	46	09BA	45	09BB	44	09BC	43	09BD	42	09BE	41	09BF	40
09C0	3F	09C1	3E	09C2	3D	09C3	3C	09C4	3B	09C5	3A	09C6	39	09C7	38
09C8	37	09C9	36	09CA	35	09CB	34	09CC	33	09CD	32	09CE	31	09CF	30
09D0	2F	09D1	2E	09D2	2D	09D3	2C	09D4	2B	09D5	2A	09D6	29	09D7	28
09D8	27	09D9	26	09DA	25	09DB	24	09DC	23	09DD	22	09DE	21	09DF	20
09E0	1F	09E1	1E	09E2	1D	09E3	1C	09E4	1B	09E5	1A	09E6	19	09E7	18
09E8	17	09E9	16	09EA	15	09EB	14	09EC	13	09ED	12	09EE	11	09EF	10
09F0	0F	09F1	0E	09F2	0D	09F3	0C	09F4	0B	09F5	0A	09F6	09	09F7	08
09F8	07	09F9	06	09FA	05	09FB	04	09FC	03	09FD	02	09FE	01	09FF	00

Ασκηση 2

```
START:
    LDA 2000H
    RRC
    JC START          ; if 1st is ON loop

STATE1:                ; 1st OFF
    LDA 2000H
    RRC
    JNC STATE1

STATE2:                ; 2nd is ON
    LDA 2000H
    RRC
    JC STATE2
    CALL DISPLAY       ; if it goes off go to Display
    JMP END

DISPLAY:
    PUSH PSW
    PUSH D
    PUSH B
    MVI D,00H          ; D is a counter, when it reaches 15 ,
                        ; the program ends.
    MVI E,00H          ; E is a flag to know if the time has
                        ; restarted (OFF-ON-OFF-ON).
    LXI B,01F4H        ; BC = 500d
```

```

CHECK1:
    MVI E,00H
    LDA 2000H
    RRC
    JNC A1          ; if the LSB is still off go to A1
                   ; if LSB = ON we go to CHECK2 and we
                   ; restart the counter (D =0).

CHECK2:
    MVI E,01H
    LDA 2000H
    RRC
    JC A1
    MVI E,00H       ; E = 0 again because we have restarted
                   ; our timer by MVI D,00H
                   ; if we didnt make E = 0 again the timer
                   ; will restart again.

    MVI D,00H

A1:
    XRA A           ; A = 00H
    STA 3000H
    CALL DELB
    CMA
    STA 3000H
    CALL DELB
    INR D
    MOV A,D
    CPI 0FH
    JZ R_END
    MOV A,E
    CPI 00H
    JZ CHECK1
    JNZ CHECK2

R_END:
    POP B
    POP D
    POP PSW
    RET

END:
    END

```

Άσκηση 3 (i)

```
START:
    LDA 2000H
    MVI E,00H
    MVI D,00H                ; D is RLC counter
                             ; D = the position of the first '1'
                             ; starting counting form MSB to LSB

    CPI 00H                  ; if the input number = 00H,
    JZ DISPLAY               ; jump to DISPLAY

BIT1:
    RLC
    INR D                    ; CY flag not affected
    JNC BIT1                 ; if bit = 0 get the next one
    MVI A,00H

LOO:
    MOV E,A
    MOV A,D
    CPI 00H
    JZ DISPLAY
    DCR D
    MOV A,E
    INR A                    ; A = A + 1
    RRC                      ; and RRC , in order to put (A7) <- (A0) .
    JMP LOO                  ; We do this D times.

DISPLAY:
    MOV A,E
    CMA
    STA 3000H
    JMP START

END
```

Άσκηση 3 (ii)

```
LXI B,01F4H

START:
    MVI D,00H           ; Counter for 4 times 5-8
    MVI E,00H           ; Counter for 4 times 1-4

    CALL KIND

    CPI 00H              ; if the input number is 0
    JZ ERROR             ; jump to ERROR

    CPI 09H              ; We will open the 1st bit (LSB LED) if
    JP ERROR             ; the input number is >= 9

                                ; When we reach this line we are sure that
                                ; the input is not 0 and that it is < 9.

    CPI 05H
    JP 5-8               ;if it is >=5 (and <9 , and not 0)
                        ; jump to 5-8
```

1-4: ; Here we are sure that our input is
; 1 <= x <= 4

MVI A, F0H
STA 3000H
CALL DELB

MVI A, FFH
STA 3000H
CALL DELB

MOV A, E
CPI 03H
JZ START

INR E
JMP 1-4

5-8:

MVI A, 0FH
STA 3000H
CALL DELB
MVI A, FFH
STA 3000H
CALL DELB

MOV A, D
CPI 03H
JZ START
INR D
JMP 5-8

ERROR:

MVI A, FEH
STA 3000H
JMP START

END

Άσκηση 4η

```
START:
    LDA 2000H
    MOV B,A                ; B holds the input

X1:
    ANI 0CH                ; isolation of bits 2,3
    CPI 00H                ; if bits 2,3 are 00 then
    JZ ZERO_OUT0          ; x0, x1 are both zero

    MVI C,02H              ; else x1 = 1 and lets see about x0

X0:
    MOV A,B
    ANI 03H                ; isolation of bits 0,1
    CPI 00H                ; if bits 0,1 are 00
    JZ X2                  ; then x0 = 0
    MVI C,03H              ; else x1 = 1 and x0 = 1 from previous check
    JMP X2

ZERO_OUT0:
    MVI C,00H              ; x0,x1 = 00

MVI D,00H

X2:
    MOV A,B
    ANI 30H                ; isolation of bits 4,5
    CPI 30H                ; if bits 4,5 are not 11
    JNZ X3                 ; then x3 = 0
    MOV A,C                ; else add 4 to A which holds x1x0
    ADI 04H                ; so that x3 = 1 and x1x0 are not affected
    MVI D,01H              ; if x3 = 1 make D = 1 to check whether
                           ; x3 and x4 are equal
                           ; If x3 = x4 then XOR gives 0, else 1
    MOV C,A                ; store the so far outcome to C

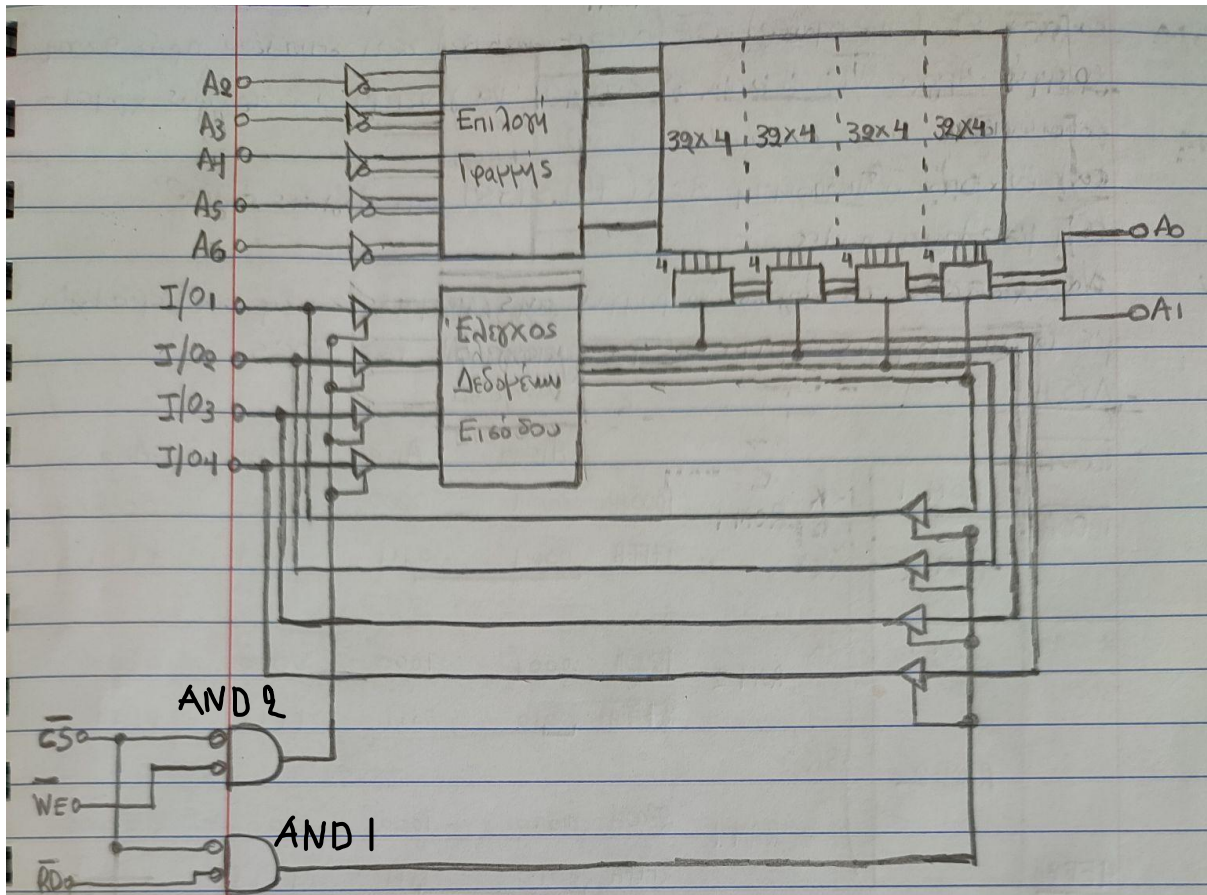
X3:
    MOV A,B
    ANI C0H                ; isolation of bits 6,7
    CPI C0H                ; check if bits 6,7 are not 11
    MVI A,00H              ; make A = 0 which does not affect flags
                           ; A will be compared to D which holds x3 value
    JNZ A                  ; if bits 6,7 are not 1,1 then AND gives 0
    MVI A,01H
```

```
A:      CMP D           ; if A = D then XOR gives 0
        MOV A,C
        JZ OUT
        ADI 08H         ; else XOR gives 1
OUT:     CMA            ; reverse logic leds
        STA 3000H       ; print the output

END
```

Θεωρητικό Μέρος

Άσκηση 5η



Έστω ότι έχουμε την διεύθυνση μνήμης 42 D = 0101010 B.

Γίνεται επιλογή γραμμής με βάση τα A2 - A6 = 01010, άρα επιλέγεται η 10η γραμμή. Όσον αφορά τις στήλες, δεδομένου ότι A1 A0 = 10 η έξοδος και των τεσσάρων πολυπλεκτών είναι η 2. Άρα, επιλέγεται η 2η στήλη σε κάθε παραλληλόγραμμο 32 x 4.

Θεωρούμε ότι CS = 1 (Αν CS = 0 απομονώνεται η είσοδος και η έξοδος).

Αν έχουμε εγγραφή:

- WE = 1, RD = 0
- Έξοδος AND1 = 0, Έξοδος AND2 = 1 άρα επιτρέπεται η ροή δεδομένων από το input προς τη μνήμη, δηλαδή επιτρέπεται μόνο η ροή μέσω των απομονωτών που βρίσκονται στα αριστερά (έτσι όπως κοιτάμε το σχήμα).

Αν έχουμε ανάγνωση:

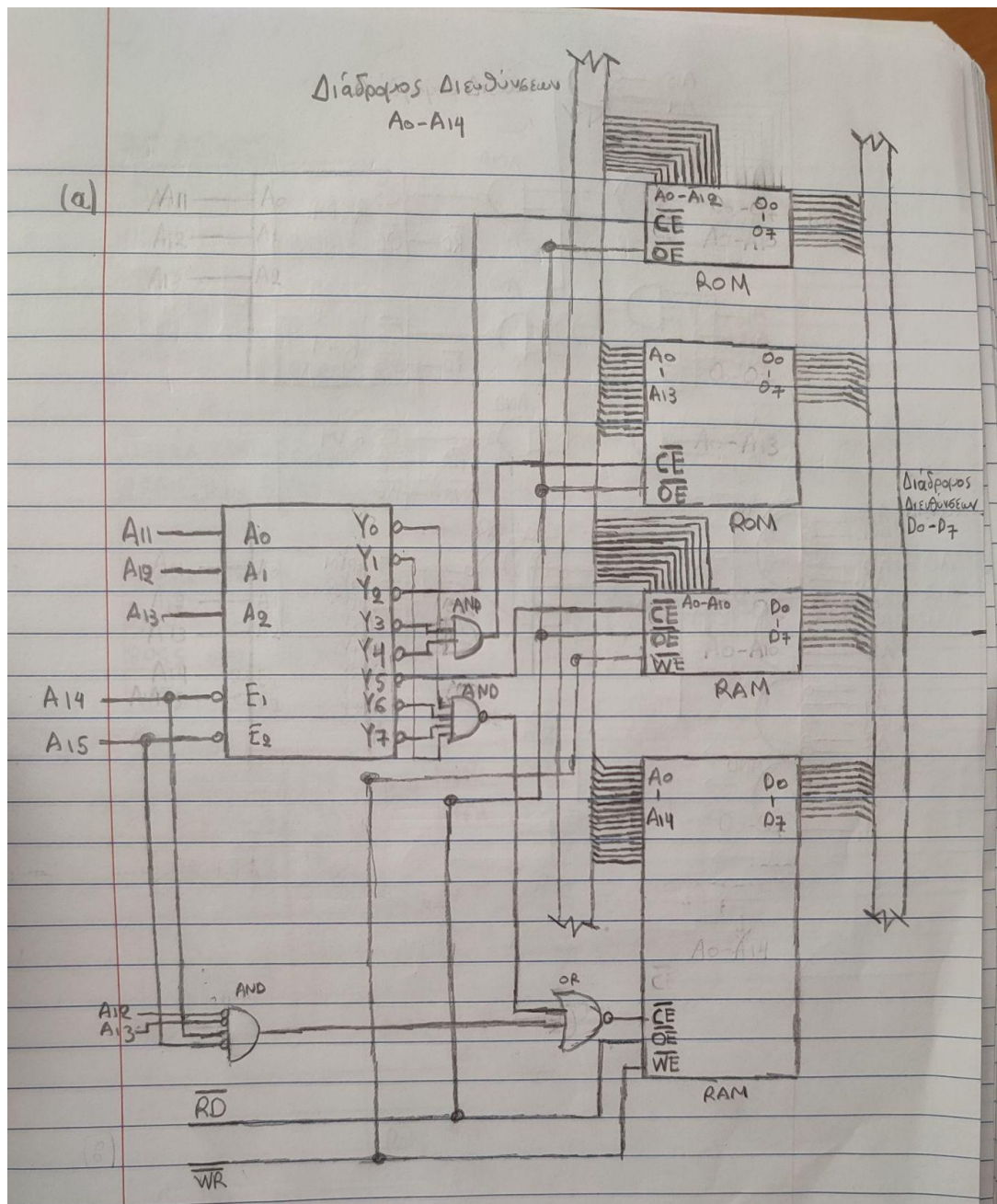
- $WE = 0, RD = 1$
- Έξοδος AND1 = 1, Έξοδος AND2 = 0 άρα επιτρέπεται η ροή δεδομένων από τη μνήμη προς το output , δηλαδή επιτρέπεται μόνο η ροή μέσω των απομονωτών που βρίσκονται στα δεξιά (έτσι όπως κοιτάμε το σχήμα).

Άσκηση 6η

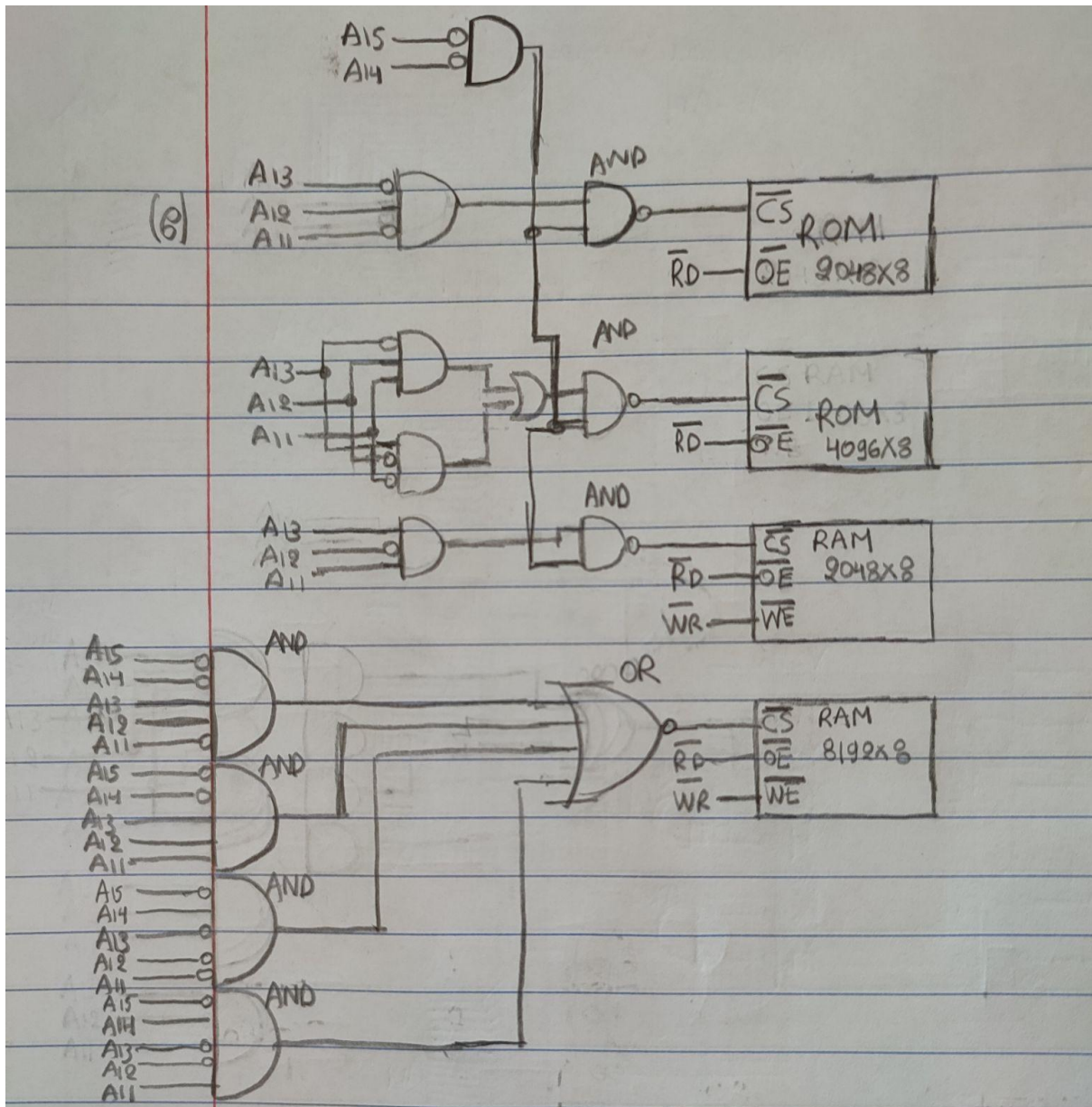
Αρχικά, απεικονίζουμε τον χάρτη μνήμης:

				A ₁₅₋₁₂	A ₁₁₋₈	A ₇₋₄	A ₃₋₀	
1000H	ROM 1	2K	ROM 1	1000H	0001	0000	0000	0000
1800H				17FFH	0001	0111	1111	1111
2800H	RAM 1	2K	RAM 2	1800H	0001	1000	0000	0000
3000H	RAM 2	8K		27FFH	0010	0111	1111	1111
				RAM 1	2800H	0010	1000	0000
4FFFH				2FFFH	0010	1111	1111	1111
			RAM 2	3000H	0011	0000	0000	0000
				4FFFH	0100	1111	1111	1111

α)



β)



7η Άσκηση

Χάρτης Μνήμης:

Address	Memory Type	Size	RAM	A15-12	A11-8	A7-4	A3-0
1000H	ROM	4K	ROM	1000H	0001	0000	0000
2000H	RAM 1		1FFFH	0001	1111	1111	1111
3000H	RAM 2	8K	RAM 1	2000H	0010	0000	0000
4000H			2FFFH	0010	1111	1111	1111
5000H	ROM	12K	RAM 2	3000H	0011	0000	0000
6000H			4FFFH	0100	1111	1111	1111
7000H			RAM	5000H	0101	0000	0000
7FFFH				7FFFH	0111	1111	1111

1000 0000 0000 0000 (8000H) : Dupa etjasa

0000 0000 1000 0000 (0030H) : Dupa etjasa

Σχεδίαση:

